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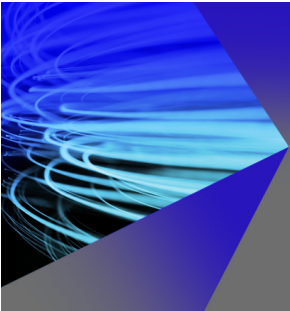
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ABSTRACT

Based on our previous visual experiments and the volume of fluid (VOF) multiphase model, the growth and detachment characteristics of a single hydrogen bubble on the horizontal microelectrode surface under the electrode-normal magnetic field have been numerically investigated. The mass transfer contributions of supersaturated dissolved hydrogen to the bubble growth from the liquid microlayer under the direct injection model and from the bulk bubble interface under the gas-liquid diffusion-controlled model are adopted. The bubble shapes and diameters predicted from the numerical investigation agree well with experimental results under the same conditions. The simulated results indicate that the supersaturated dissolved hydrogen concentration and the mass transfer source at the wedge-shaped areas adjacent to the bubble foot are obviously higher than those in the wider bulk bubble interface regions. The mass transfer contribution to the bubble growth from the liquid microlayer beneath the bubble base directly plays a dominant role. The higher current density and corresponding Lorentz force mainly appears in the wedge-shaped areas, while the higher rotational electrolyte flow velocity appear at oblique positions of the bubble equator. The bubble detachment behavior makes the rotational electrolyte flows is significantly more complex.

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I. INTRODUCTION

Hydrogen is an alternative energy source with high efficiency and pollution free. Developing the hydrogen economy has increasingly shown its importance in achieving the goals of peak carbon dioxide emissions and carbon neutrality. Water electrolysis technology plays a vital role in hydrogen production due to mature technology, abundant raw materials, clean energy, and low carbon technology, which is the subject of growing concern and criticism. Water electrolysis is a typical electrochemical gas evolution process occurring at the bubble/electrolyte/electrode multiphase interface. Hydrogen bubble growth and detachment behaviors on the electrode surface significantly influence water electrolysis efficiency,¹ which can induce local heat and mass transfer. Meanwhile, the continuous bubble adhesion reduces electrode activation area, which would, in turn, hinder the electron or ionic transmission and raises higher overpotential.^{2,3} Therefore, the comprehensive study of complex bubble evolution behavior plays an important role in revealing bubble growth characteristic and mechanism in water electrolysis.

To approach the bubble evolution behavior in the hydrogen-evolution systems, scholars have used high speed camera visualization experiments, electrochemical characterizations and measurements, and theoretical analysis to investigate the bubble nucleation, growth, and detachment. According to the numerous previous studies,^{4–7} the hydrogen bubble growth behavior was closely associated with the mass transfer of dissolved hydrogen near electrodes. Also, some typical bubble growth models were established as $d(t) = \beta t^x$, where β stands for growth coefficient and x represents time coefficient. The choose value of two parameters usually depends on the relative size of the bubbles and the electrodes. When the electrode size (large electrode) is significantly larger than the bubble size, a gas-liquid diffusion-controlled model can certainly be recognized as the dominant role, in which the bubble growth is largely due to the mass transfer of the supersaturated dissolved hydrogen from the bulk electrolyte to the hydrogen bubbles through the bubble interface. Instead, when using a very tiny electrode size (often called microelectrode), a direct injection model becomes the dominant behavior, in which the bubble hydrogen

growth is mainly caused by the mass transfer of the supersaturated dissolved hydrogen from the liquid microlayer beneath the bubble base directly. Liu *et al.*⁸ investigated the hydrogen bubble growth on the disc-shaped large electrode driven by the gas–liquid diffusion-controlled model. Actually, in many cases, the relative magnitudes of the bubble and electrode sizes change as the bubble continues to grow, and therefore, the above two mass transfer models often exist at the same time. The obvious difficulty is that the contribution of the mass transfer of the supersaturated dissolved hydrogen to the bubble growth through the liquid microlayer or through the bubble interface is difficult to ascertain. The deeper reason is the lack of characterization and geometrical structures of the liquid microlayer and the corresponding mass transfer mathematical model.

Many previous research have identified that it is very difficult to accurately track simultaneously the evolution behavior of individual bubble among the many hydrogen bubbles on the large electrodes in water electrolysis experiments. By adopting the microelectrode experiment investigations,^{9–12} researchers have effectively obtained a great amount of common bubble behaviors including single bubble growth and detachment, on the basis of which it may be possible to reveal some interesting findings such as the structure and formation characteristics of the liquid microlayer. Bashkatov *et al.*¹³ revealed that a liquid microlayer (or a microbubble carpet) accompanied by coalescence with many tiny bubbles under a main hydrogen bubble on the microelectrode surface and its thickness and variations was measured. Chen *et al.*¹⁴ proposed a simple model for predicting the liquid microlayer based on the wetting dynamics theory of the three-phase interface. We are relatively lack of the research on more details of the liquid microlayer and their quantitative predictions. Certainly, the existing investigations have provided good insight into the internal mechanisms of single bubble growth behavior on the microelectrode surface. However, due to the complexity of the liquid microlayer, relevant experimental studies are still far from adequate, especially for the mass transfer mechanism dominated by the direct injection model for the microelectrode systems.

With the advancements in numerical techniques and computing power, various numerical methods have been used to study multiphase flows and interfacial heat mass transfer.^{15–18} Most of the previous studies were the focus on the steady physical and mathematical model of a fixed bubble diameter and conducted multiple steady simulation cases with different bubble sizes in different growth instances to approximately describe the actual unsteady bubble growth process in water electrolysis.^{9–13,19,20} However, the fixed bubble diameter model only assumed the hydrogen bubble as a solid ball and was still limited to single electrolyte phase flow simulations, which cannot reasonably represent the complex gas–liquid two phase flows and the continual bubble growth process controlled by the mass transfer of supersaturated dissolved hydrogen from the bubble interface or the liquid microlayer. Therefore, the volume of fluid (VOF) method is one of the most common methods and the recommended approach for tracking multiphase interfaces.^{21–25} So far, only Liu *et al.*⁸ and Vachaparambil *et al.*^{26,27} has been numerically simulated a single hydrogen bubble growth behavior and the mass transfer in a large electrode system only considering gas–liquid diffusion-controlled model through bubble interface using the VOF method. As explained earlier, the direct injection model through the liquid microlayer may also happen interactively in the whole bubble growth process. Not only that, due to the

difficulty in large electrode measurements, the available microelectrode experiments can provide more detailed and more information of a single bubble growth, which can further provide a coherent framework for modeling the mass transfer dynamics. Overall, relevant theoretical research and modeling method on the bubble growth behavior and the mass transfer during the hydrogen-evolution systems are rare and it should take deep research. Fortunately, many similar bubble physical phenomena and behaviors exist during hydrogen bubble growth on the electrode surface and vapor bubble growth on the boiling surface. Moreover, many research studies have been carried out to study single vapor bubble growth and mass transfer involved in boiling heat transfer systems by VOF simulation and experiments.^{28–31}

Specifically, applying an easy-operation, simple, and inexpensive external electrode-normal magnetic field can generate microscopic magnetohydrodynamic (Micro-MHD) convection or the rotational flows due to the local electric lines bend around a single insulation bubble on the microelectrode surface.^{9–12,20,32} Our previous experimental investigations^{32,33} revealed that the Micro-MHD effects on the growth and detachment behavior of a single bubble on both the horizontal and vertical microelectrodes were closely related to the magnetic field and current density conditions. However, the research on the single bubble growth and the above-mentioned two mass transfer behaviors and their models on the microelectrodes at the electrode-normal magnetic fields has not been carried out systematically.

Therefore, to better understand the bubble growth mechanisms on the horizontal microelectrode surface under electrode-normal magnetic fields in water electrolysis, the VOF method was used to capture the bubble interface and simulate the hydrogen bubble growth in a two-dimensional (2D) axisymmetric region. The coupling mass transfer of the supersaturated dissolved hydrogen between the gas–liquid diffusion-controlled model through the bubble interface and the direct injection model through the liquid microlayer beneath the bubble was considered. The numerical model was validated by comparison with our previous experimental data. Finally, the bubble growth and mass transfer characteristics and spatial multi-fields distributions around a single hydrogen bubble were obtained.

II. MATHEMATICAL AND PHYSICAL MODELS

A. Governing equations

In the VOF method, the gas–liquid two-phases presenting a control volume share a set of momentum equations, which is suitable for the modeling of the two immiscible and laminar fluid flows with the clear phase interface. For the single hydrogen bubble growth on the microelectrode surface, there is an obvious interface between the hydrogen bubble and the electrolyte. In the VOF model, the geometric reconstruction scheme that is based on the piece linear interface calculation (PLIC) method³⁴ is applied to accurately capture and obtain the clear bubble interface during the bubble dynamic growth process. In the VOF model, the volume fraction of the liquid α_k is presented. Within each computational cell, $\alpha_k = 1$ means that there is filled with only liquid phase and $\alpha_k = 0$ means that there is no liquid phase. When α_k is between 0 and 1, there is a mixture of liquid and gas phases in the computational cell, which represents the gas–liquid interface.

In the current VOF model, the primary phase is the electrolyte and the hydrogen gas is the secondary phase. Based on the resolution of the volume fraction field, the continuity equations can be written as follows:

$$\frac{\partial \alpha_k}{\partial t} + \nabla \cdot (\alpha_k \mathbf{u}) = \frac{S_m}{\rho_k}, \quad (1)$$

where $\mathbf{u}$ is the velocity vector. The volume fraction sum of the two phases in any computational cell is equal to unity ($\alpha_g + \alpha_l = 1$). The subscripts g and l represent the hydrogen gas phase and electrolyte phase, respectively. The mass source term S_m represents the mass transport of the supersaturated dissolved hydrogen both from the liquid microlayer under the direct injection model and from the bulk bubble interface under the gas–liquid diffusion-controlled model, resulting in the dynamic hydrogen bubble growth performance.

A single momentum equation is solved throughout the domain and the velocity field is shared between the two phases, which can be written as follows:

$$\frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla [\mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T)] + \rho \mathbf{g} + \mathbf{F}_\sigma + \mathbf{F}_L, \quad (2)$$

where $\rho = \alpha_g \rho_g + \alpha_l \rho_l$ and $\mu = \alpha_g \mu_g + \alpha_l \mu_l$ are the mixture density and viscosity related to the volume fraction, density, and viscosity for each phase in the computational cell. $\mathbf{F}_\sigma$ represents the surface tension force per unit volume, which is calculated by applying the continuum surface force (CSF) model.³⁵ $\mathbf{F}_L$ represents the Lorentz force induced by the orthogonal interaction of the electric and magnetic fields around an insulating hydrogen bubble, which will be covered in more detail later. In the CSF model, the surface tension force is modeled as the form of a body force, which is implemented into the momentum equation source. The formula for $\mathbf{F}_\sigma$ is calculated as follows:

$$\mathbf{F}_\sigma = \sigma \frac{\rho \kappa \nabla \alpha_l}{0.5(\rho_l + \rho_g)}, \quad (3)$$

where σ is the surface tension coefficient. κ is the local phase interface curvature, which is calculated in terms of the unit interface normal $\kappa = \nabla \cdot \hat{n} = \nabla \cdot (\Delta \alpha_l / |\nabla \alpha_l|)$. $\hat{n}$ is the unit interface normal vector, which is further calculated from the gradient of the volume fraction.

In the CSF model, the unit vector of the computational cell between the hydrogen bubble and the microelectrode can be calculated by the following formula:

$$\hat{n} = \hat{n}_w \cos \theta_w + \hat{t}_w \sin \theta_w, \quad (4)$$

where $\hat{n}_w$ and $\hat{t}_w$ are the normal and tangential unit vectors of the wall, respectively. θ_w is the contact angle between the hydrogen bubble and the microelectrode wall. This study makes use of our previous available experimental data to determine the dynamic contact angles, which will be described next.

B. Electric field calculation model

The hydrogen bubble is the insulating phase and the electrolyte is the conductive phase. The potential distribution in the electrolyte can be calculated from the potential equation as follows:

$$\nabla \cdot (\kappa_m \nabla \phi) = 0, \quad (5)$$

where κ_m and ϕ are the mixture conductivity of the electrolyte and the internal cell potential, respectively. Due to the presence of the dissolved hydrogen components in the electrolyte phase, κ_m can be calculated according to Maxwell relations,³⁶

$$\kappa_m = \kappa \frac{1 - \alpha_g}{1 + 0.5\alpha_g}, \quad (6)$$

where κ is the conductivity of the pure electrolyte and α_g is the local gas volume fraction.

The electric field distribution can be obtained by solving the user defined scalar (UDS) equation,

$$\frac{\partial(\rho \phi_k)}{\partial t} + \frac{\partial}{\partial x_i} \left(\rho u_i \phi_k - \Gamma_k \frac{\partial \phi_k}{\partial x_i} \right) = S_{\phi_k}, \quad (7)$$

where ϕ_k is the potential scalar, Γ_k is the diffusion coefficient, and S_{ϕ_k} is the scalar source, respectively. The second term is zero. Obviously, the diffusion coefficient Γ_k just equals the mixture conductivity κ_m .

The current density distribution is closely correlated with the potential distribution in the cell, which can be deduced from Ohm's law,

$$\mathbf{j} = \kappa_m \nabla \phi, \quad (8)$$

where $\mathbf{j}$ is the current density vector and $\mathbf{B}$ is the magnetic field vector, respectively. The Lorentz force $\mathbf{F}_L$ can be calculated as $\mathbf{F}_L = \mathbf{j} \times \mathbf{B}$ and will be added into the momentum equation in the form of body force source.

C. Dissolved hydrogen transport model

The total amount of the charge through the working electrode (microelectrode) surface determines the generation rate of the hydrogen bubbles under the constant current electrolysis condition. According to the principle of charge conservation, the generation rate of the hydrogen component on the microelectrode surface can be calculated by Faraday's law,

$$m_{H_2} = \frac{M_{H_2} I}{2FA_e}, \quad (9)$$

where m_{H_2} is the generation rate of the hydrogen components. M_{H_2} is the molar mass of hydrogen. F is the Faraday constant. I is the electrolysis current. A_e is the microelectrode area.

In order to simplify the mathematical model, the existence of other ions in the electrolyte is ignored and only the transportation process of the dissolved hydrogen component is considered. The diffusion and transport equation of the dissolved hydrogen components can be calculated as follows:

$$\frac{\partial}{\partial t}(\rho_l Y_{H_2}) + \nabla \cdot (\rho_l \mathbf{u} Y_{H_2} - D_{H_2} \nabla Y_{H_2}) = S_{H_2}, \quad (10)$$

where Y_{H_2} and D_{H_2} are the mass fraction (kg/kg) and the diffusion coefficient of the dissolved hydrogen component in the electrolyte, respectively. The generation rate of the hydrogen components can be regarded as the diffusion equation source S_{H_2} .

D. Mass transfer models

Due to the continuous electrochemical reaction, the dissolved hydrogen component can accumulate on the electrode surface. Therefore, the dissolved hydrogen component concentration can easily reach the supersaturated dissolved hydrogen concentration conditions. The supersaturated concentration difference at the bubble

interface acts an important driving force for the interfacial mass transport from the dissolved hydrogen component in the bulk electrolyte to the hydrogen bubble. The interfacial mass transfer rate at the gas–liquid interface can be calculated,

$$m_i = k_{H_2} M_{H_2SO_4} (c_b - c_s) = k_{H_2} \rho_l (Y_{H_2} - Y_{s,H_2}), \quad (11)$$

where m_i is the interfacial mass transfer rate. $M_{H_2SO_4}$ is the molar mass of electrolyte. k_{H_2} is the interfacial mass transfer coefficient. c_0 and c_s are the bulk molar concentration and saturation molar concentration of the dissolved hydrogen component in the electrolyte, respectively. Y_{H_2} and Y_{s,H_2} are the corresponding bulk mass concentration and saturation mass concentration of the dissolved hydrogen components, respectively.

For the dissolved hydrogen components transport in the cells, k_{H_2} can be calculated by the following formula:³⁷

$$k_{H_2} = \frac{D_{H_2}}{2r_b} \left(2 + 0.6 \left(\frac{2r_b v_l \rho_l}{\mu_l} \right)^{1/2} \left(\frac{\mu_l}{\rho_l D_{H_2}} \right)^{1/3} \right) (c_b - c_s), \quad (12)$$

where r_b is the bubble dynamic growth radius. v_l is the electrolyte characteristic velocity and can be calculated as $v_l = \sqrt{(dr_b/dt)^2 + w^2}$. w is the rotational electrolyte velocity around the single bubble caused by Lorenz force under the electrode-normal magnetic fields in water electrolysis.

The microelectrode area is very small, and therefore, the current density is very large. The mass transfer process of the supersaturated dissolved hydrogen driving the bubble growth mainly occurs near the bubble foot area for the microelectrode electrolysis systems. Yang *et al.*¹² compared the actual generated hydrogen bubble volume V with the ideal bubble volume V_{EC} calculated by the charge amount in the horizontal microelectrode experiments and found that the ratio V/V_{EC} was close to 1. During the whole single bubble growth cycle, the gas absorption efficiency on the microelectrode surface was usual higher than 0.94. The mass transfer rate of the direct injection model through the liquid microlayer beneath the bubble base is much higher than that of the gas–liquid diffusion-controlled model through bubble interface. A circular region with the radius of $2d_e$ is divided at the contact position between the bubble base and the microelectrode surface. The direct injection model through the liquid microlayer causing the bubble growth was calculated and incorporated into the VOF simulation by adding mass transfer source terms via user-defined functions (UDFs), which would be described in more detail below.

E. Transformation approach between area and volume mass transfers

The generation rate of hydrogen component m_{H_2} and the mass transfer rate m_i at the bubble base and bulk gas–liquid interface regions are both the typical mass transfer rates per unit area. For the VOF model, the mass transfer rate per unit area should be converted into the mass transfer rate per unit volume during the simulation of mass transfer.

For the mass transfer in the liquid microlayer, the dissolved hydrogen components generated on the microelectrode surface were regarded as the direct diffusion sources. During each iteration process, the UDF programs loaded the mass source term to the first computational cell adjacent to the microelectrode surface, which was

considered as the effective hydrogen generation rate. The volume mass transfer rate of the dissolved hydrogen component in the first computational cell can be calculated as follows:

$$m_{H_2,V} = \frac{A_e m_{H_2}}{V_c} = \frac{m_{H_2}}{L_c}, \quad (13)$$

where V_c is the volume of the first computational cell. L_c is the height of the first computational cell.

For the mass transfer process at the bulk bubble interface, the area mass transfer rate of dissolved hydrogen components m_i can be converted into the volume mass transfer rate $m_{i,V}$ through the bubble surface area A_b and the bubble interface volume V_i . Due to the small bubble volume on the microelectrode surface during water electrolysis, the bubbles can be approximately regarded as spherical, which can be also observed from our previous experiments³² and other similar experimental observations.^{10–13} According to the bubble contact angle θ_w and bubble diameter d_b , the bubble surface area A_b can be calculated as follows:³⁷

$$A_b = \frac{\pi}{2} (1 + \cos \theta_w) d_b^2. \quad (14)$$

The bubble interface volume can be obtained by the accumulative volumes of the mesh element $V_{c,i,l}$ and the electrolyte volume function $\alpha_{c,i,l}$ in each mesh element at the bubble interface,

$$V_i = \sum_{i=1}^n \alpha_{c,i,l} \cdot V_{c,i} \quad (0.1 < \alpha_{c,i,l} < 0.9). \quad (15)$$

The electrolyte volume function in the range of 0.1–0.9 is defined as the computed gas–liquid interface. n represents the total number of the meshes in the gas–liquid interface. The volumetric mass transfer rate at the bubble interface can be calculated as follows:

$$m_{i,V} = \frac{A_b}{V_i} m_i. \quad (16)$$

F. Computational domain

Based on the single bubble growth characteristics on the microelectrode surface obtained by our previous visualization water electrolysis experiments,³² the three-dimensional (3D) hydrogen bubble can apply to spherical shape with axial rotating symmetry. Therefore, a two-dimensional (2D) axisymmetric model was adopted, and the x -axis was set as the symmetry axis for reducing the computational cost. The gravitational acceleration is imposed in the axisymmetric direction. The 2D computational domain is a rectangular domain of $1.8 \times 3.6 \text{ mm}^2$, as shown in Fig. 1. It is considered that the bubble diameter is relatively small compared to the size of water electrolysis cell, and therefore the effect of the cell wall is neglected, respectively.

The virtual 3D rotational axisymmetric geometric model of the computational domain is shown in Fig. 1(a). The top of the computing domain is set as the anode. A virtual ring cathode was used to replace the microelectrode and was created to represent the three-phase contact line at the bubble bottom [the red part in Fig. 1(a)]. The cathode inner diameter d_{in} is equal to the bubble contact diameter d_c . Thus, the total amount of the current through the ring cathode surface remains unchanged if the current density is fixed. The ring cathode area was equal to the microelectrode area A_e . The ring cathode outer diameter

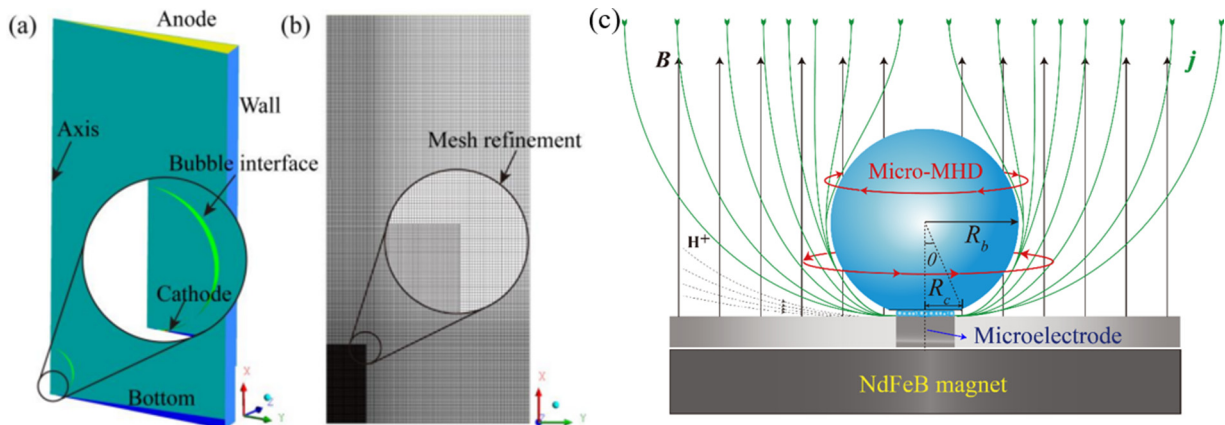


FIG. 1. Computational domain, meshing and micro-MHD convection (a) computational domain and boundary conditions, (b) local mesh refinement, and (c) detailed micro-MHD convection around a single bubble on the horizontal microelectrode surface.

d_{out} changes with the bubble contact diameter d_c during the continuous bubble growth, which was obtained as $d_{out} = \sqrt{4A_e/\pi + d_m^2}$. The actual 2D computational domain is shown in Fig. 1(b). To clearly display the bubble interface and accurately capture the bubble growth, the mesh refinement and the mesh sensitivity testing for VOF simulation were considered carefully in the bubble growth location region of $0.7 \times 0.7 \text{ mm}^2$. The high Lorentz forces are generated close to the bubble due to the high local current density, as shown in Fig. 1(c). The black line with an arrow indicates the magnetic field perpendicular to the horizontal microelectrode. The blue line shows the current line distribution around a single bubble. The red arrow line represents the Micro-MHD convection or rotational electrolyte flow generated around a growing bubble on the horizontal microelectrode.

G. Bubble contact diameter and dynamic contact angle

The bubble contact angle θ_w changes with the bubble diameter during the single bubble growth. It is an important parameter to

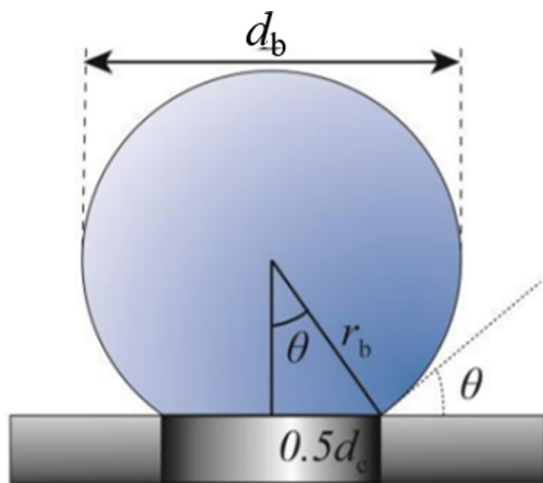


FIG. 2. Schematic of bubble contact angle.

accurately predict the bubble growth and detachment process. It was found that there was a certain relationship [that is, $\theta_w = \arcsin(d_c/d_b)$] between the bubble diameter d_b , the bubble contact diameter d_c and the bubble contact angle θ_w , as presented in Fig. 2. Based on the experimental data and analysis results, a quadratic polynomial model is proposed to describe the non-linear relationship between the bubble diameter d_b and the bubble contact diameter d_c with open-source machine learning in Python framework scikit-Learn,³⁸

$$d_c = \beta_0 + \beta_1 d_b + \beta_2 d_b^2, \quad (17)$$

where β_0 , β_1 , and β_2 are the coefficients of the different terms, respectively.

A total of 29 100 experimental results about the bubble diameter and the bubble contact diameter in 72 bubble growth cycles were selected in the polynomial regression data set to get the training of regression models. After the training, the bubble diameter in the test d_{test} was brought into the quadratic polynomial to calculate the predicted value of bubble contact diameter $d_{c,pred}$. Then, $d_{c,pred}$ was

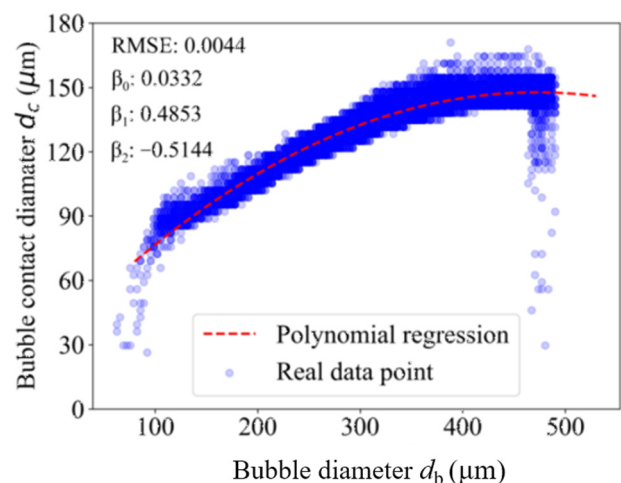


FIG. 3. The variations of bubble contact diameter with bubble diameter.

compared with the actual bubble contact diameter $d_{c,\text{test}}$ in the original test set. The root mean square error (RMSE) was used as the polynomial regression evaluation index. The test results showed that the RMSE is only 0.0044 mm and the prediction accuracy is satisfied,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (d_{c,\text{test}} - d_{c,\text{pred}})^2}. \quad (18)$$

The variations of the bubble contact diameter with the bubble growth diameter have been investigated in Fig. 3. It can be observed from Fig. 3 that the bubble contact diameter is larger than the microelectrode diameter ($100 \mu\text{m}$) used in our previous experiments³² for most of bubble growth time. However, in such circumstances, the microelectrode activation area would smaller than the bubble contact area, but the hydrogen-evolution process still happens and the hydrogen bubble continues to grow. The result has proven that certain liquid

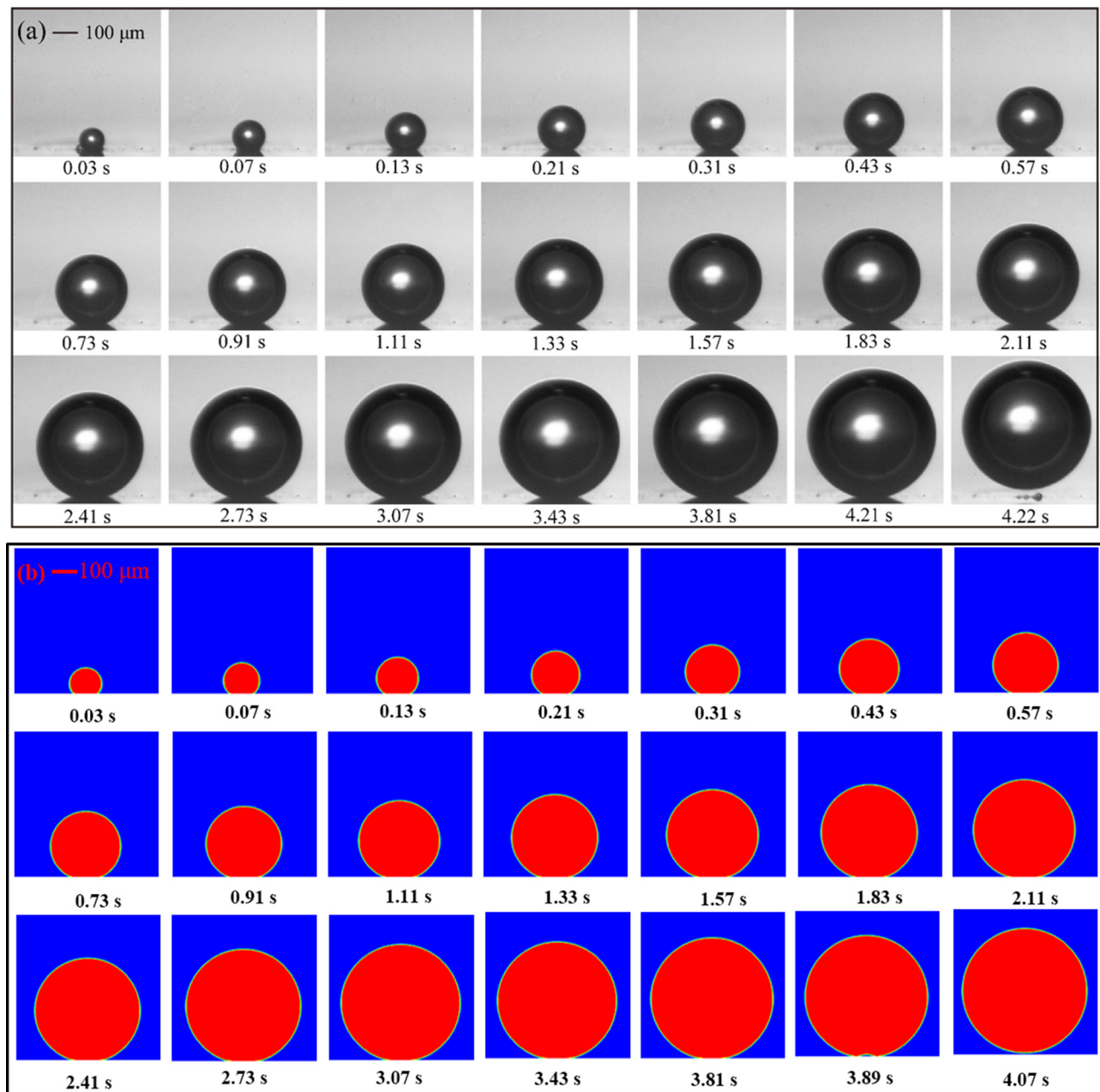


FIG. 4. Comparison of numerical bubble growth evolution with experimental results on the horizontal microelectrode surface: (a) experimental observations and (b) numerical results.

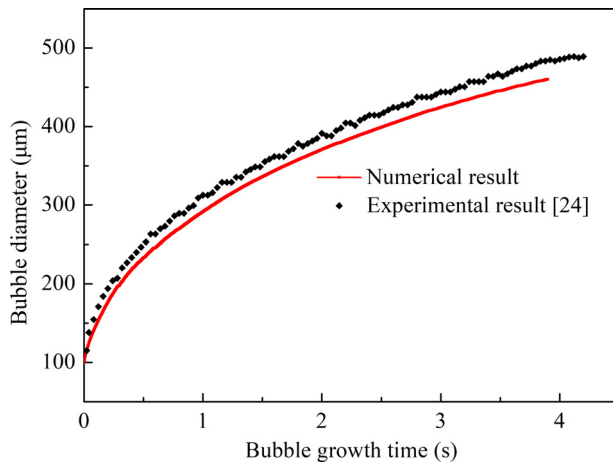


FIG. 5. Bubble diameter evolution during a single growth cycle in experiment and simulation results.

microlayer beneath the bubble base does exist and it thus leads to the electrochemical reaction at the microelectrodes. The relationship between the bubble contact diameter and bubble diameter is quadratic and non-linear. When the bubble diameter is known, the exact value

of the bubble contact diameter can be predicted. On that basis, the dynamic bubble contact angle θ_w can be obtained according to the relationship $\theta_w = \arcsin(d_c/d_b)$,

$$d_c = 0.0332 + 0.4853d_b - 0.5144d_b^2. \quad (19)$$

H. Boundary conditions and numerical method

The VOF simulations were set up to closely match the experimental conditions (current density with 1.5 A/cm^2 and electrode-normal magnetic field intensity with 415 mT). An initial spherical bubble nucleus with a bubble diameter of $100 \mu\text{m}$ and a bubble contact diameter of $76.6 \mu\text{m}$ was at the center of the axisymmetric model. The distance of the bubble center point from the microelectrode surface was $62.9 \mu\text{m}$. The anode was set as the pressure-outlet boundary condition, and the anode potential is 0 V . The no-slip wall boundary condition was applied at the bottom and the right side of the calculation domain. The momentum, potential, and dissolved hydrogen transport equations were solved using a second-order upwind difference scheme, while the geometric reconstruction method was used to solve the volume fraction equation. The pressure-velocity coupling was performed with a pressure implicit splitting of operators (PISO) algorithm. The densities of hydrogen gas phase and electrolyte phase are 0.0899 and 998.2 kg/m^3 , respectively. The conductivity of the pure electrolyte is

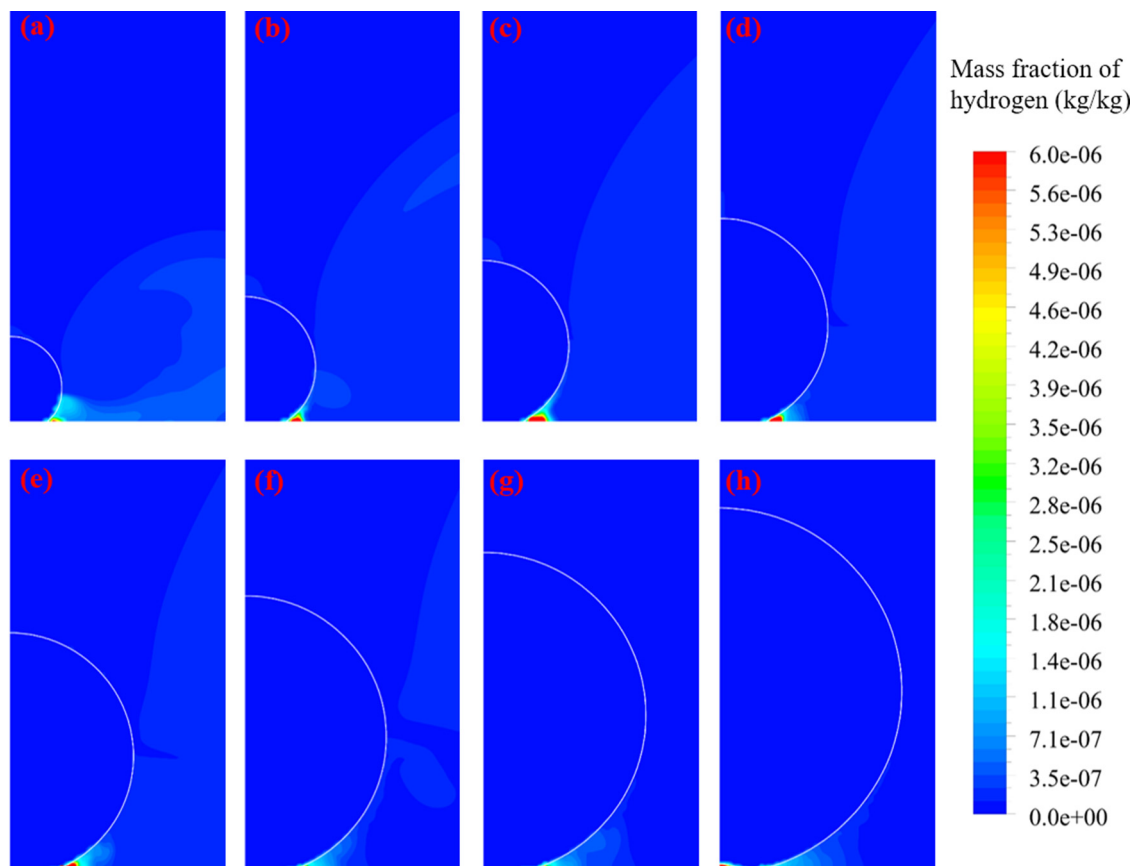


FIG. 6. Dissolved hydrogen concentration on the horizontal microelectrode surface at different time instances (a) 0.05; (b) 0.2; (c) 0.4; (d) 0.8; (e) 1.2; (f) 1.8; (g) 2.7; and (h) 3.89 s.

32.5 S/m. The saturation mass concentration of the dissolved hydrogen components is 1.5×10^{-6} kg/kg.

III. RESULT AND DISCUSSION

A. Model validation

In order to evaluate and validate the proposed VOF approach, the comparison of the dynamic growth images of a single hydrogen bubble on the horizontal microelectrode surface during a growth cycle between the experimental and the simulated results are shown in Fig. 4. The gas–liquid interface in the simulation is reconstructed as the volume fraction isosurface of 0.5 as presented in Fig. 4(b). Both the bubble images in the simulation and experiments for different time instances are drawn on a scale of the same uniform size. Clearly, the observation indicates that the simulated and experimental hydrogen bubbles are spherical and there is reasonable agreement between them.

For further quantitative validation, Fig. 5 compares the simulated bubble diameter evolutions in one bubble growth cycle with the experimental data. The plot shows that the hydrogen bubble grows very fast and the bubble growth rate is large at the initial growth stage, and then the bubble growth rate decreases gradually until it departs from the microelectrode surface. Overall, satisfactory qualitative and quantitative agreement is observed between the simulated and experimental bubble shapes and diameters during the whole bubble growth process. The

simulated bubble diameter is a little smaller than the experimental value. The bubble growth time and bubble detachment diameter obtained by VOF simulations are also a little bit smaller than those obtained from the experiments. There are some small differences between the numerical and experimental results because the possible coalescence behavior of many tiny bubbles at the bottom of the single main hydrogen bubble is ignored. In addition, the OpenCV video image processing program proposed in our previous experiments started to record very clear images of bubble diameter information when the main bubble appears on the microelectrode surface. The result is that the corresponding moment from the start records the main bubble diameter information in experiments, which is not exactly at $t = 0$ but after an initial nucleation time. The nucleation time is very short, and it is ignored in the numerical simulation, which may also cause some small differences compared with the experimental results. These results confirm that the VOF simulation results and calculation method in this study are highly effective and reasonable.

B. Hydrogen concentration distribution and interfacial mass transfer

As previously mentioned, the mass transfer causing a single hydrogen bubble growth is mainly determined by the supersaturated

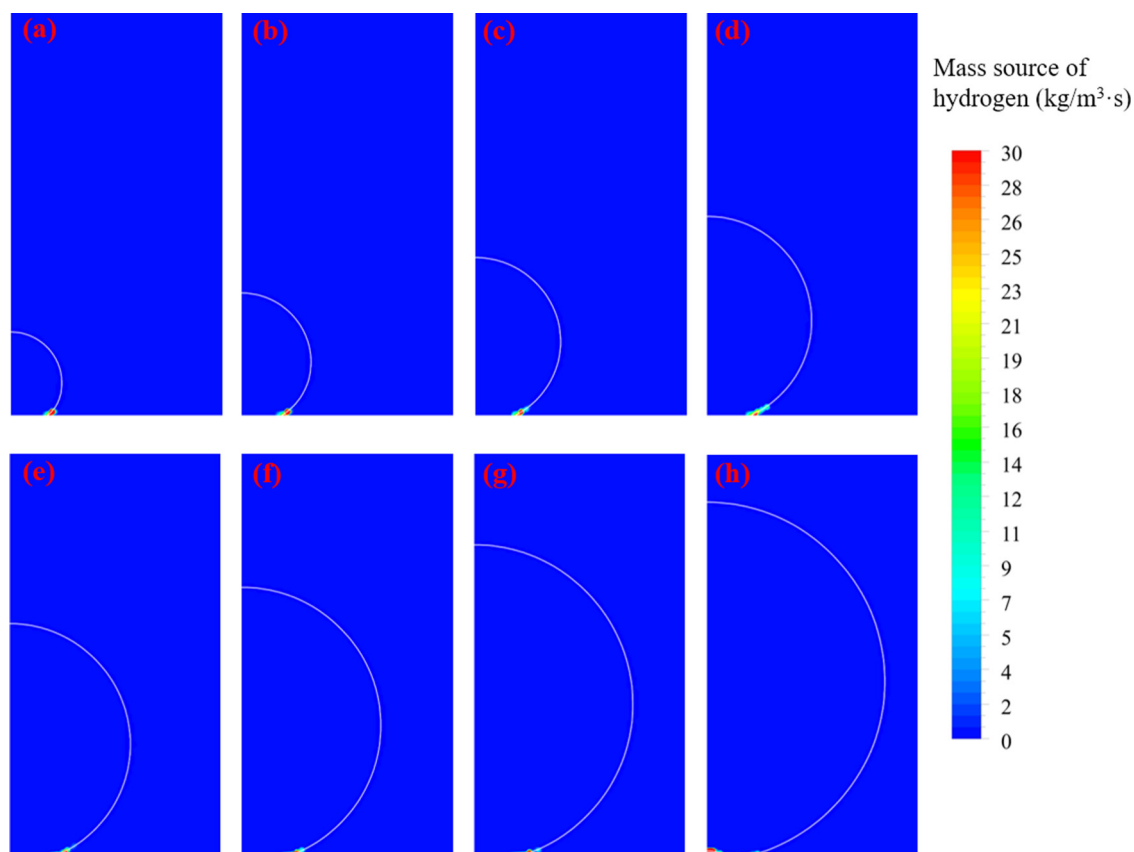


FIG. 7. Mass transfer source of dissolved hydrogen at the gas–liquid interface at different time instances: (a) 0.05s, (b) 0.2, (c) 0.4, (d) 0.8, (e) 1.2, (f) 1.8, (g) 2.7, and (h) 3.89 s.

concentration conditions of the dissolved hydrogen components near the bubble interface regions. The evolutions of the dissolved hydrogen concentration (mass fraction, kg/kg) distribution around a single hydrogen bubble on the horizontal microelectrode surface at different time instances are presented in Fig. 6. In addition, the corresponding interfacial mass transfer source distributions of the dissolved hydrogen at each time instance are illustrated in Fig. 7. The white line indicates the spherical bubble profile interface. In general, the dissolved hydrogen concentration at the wedge-shaped areas adjacent to the bubble foot appears significantly higher than that in the wider bulk electrolyte regions and presents the obvious concentration gradient. Thus, a closer examination shows that the dissolved supersaturated hydrogen concentration condition (approximately larger than saturation concentration of 1.5×10^{-6} kg/kg) exists mainly in the wedge-shaped areas. As the electrolysis process continues, the bubble volume becomes larger, and the supersaturated hydrogen concentration area gradually becomes larger. As mentioned earlier, since the supersaturated hydrogen concentration mainly exists in the wedge-shaped areas, the mass transfer source of dissolved hydrogen is also large as expected and it is almost zero in most of the bulk bubble interface regions, as indicated in Fig. 7. The value of the hydrogen mass source is closely related to the dissolved hydrogen concentration field. Therefore, it can be concluded that the dissolved hydrogen mass transfer process from

the liquid microlayer beneath the bubble base directly plays a dominant role. By comparison, the mass transfer rate of the supersaturated dissolved hydrogen from the bulk electrolyte to the bubbles through the bubble interface is rather small. This study can confirm that the mass transfer characteristics of the supersaturated dissolved hydrogen are in accordance with the explanation theory.

C. Distribution of current density and Lorentz force

Based on the VOF interface tracking method, the dynamic evolutions of the current density and the Lorentz force distributions around a single hydrogen bubble are simulated during the bubble growth. Figure 8 shows the distributions of the current line and the corresponding x -component of current density around a hydrogen bubble interface at some typical bubble growth time instances. The simulation results show that all the current lines start from the large scale anode surface to the tiny microelectrode surface eventually. The current lines present significantly distorted as they flow across the insulated bubble surface and different current parts are linked together through such a tiny microelectrode area, resulting in a high current density distribution at the wedge-shaped areas of the bubble foot and a very low current density in most of the bulk electrolyte regions. The distortion and

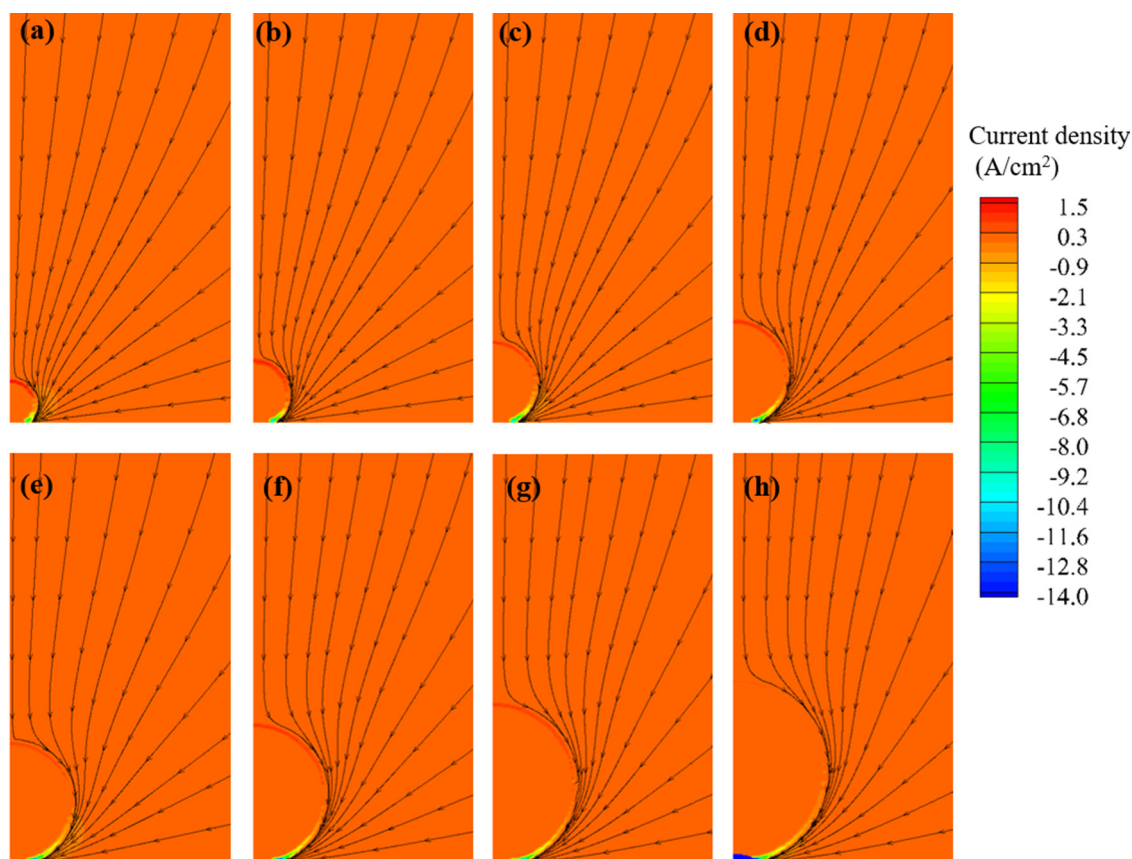


FIG. 8. Current distribution around a single hydrogen bubble on the horizontal microelectrode surface at different time instances: (a) 0.05, (b) 0.2, (c) 0.4, (d) 0.8, (e) 1.2, (f) 1.8, (g) 2.7, and (h) 3.89 s.

accumulation effects of the current lines at the wedge-shaped areas become more pronounced, and the absolute value of the current density also increases as the bubble diameter increases. We were amazed to find that the bubble can only make contact with part portion of the microelectrode when the bubble just detaches from the horizontal microelectrode surface at about 3.89 s. This would cause some change of the electric field distribution. The simulated magnitude and direction of the current density may significantly change at the bubble bottom as presented in Fig. 8(h).

The simulated Lorentz forces distribution around a single hydrogen bubble interface at some typical bubble growth time instances are presented in Fig. 9. The Lorentz force magnitude distributions in the horizontal direction of rotation generated around a single hydrogen bubble are mainly determined by the current density and electrode-normal magnetic field distribution characteristics. Although the microelectrode surface boundary was given a uniform magnetic field intensity with $B = 415$ mT, B can reduce quickly along wall-normal distance. Because the current density at the wedge-shaped areas is very high as previously confirmed in Fig. 8, the higher value of the generated Lorentz force is mainly concentrated at the wedge-shaped areas. The Lorentz force points vertically into the picture, and we take this direction as positive from the two-dimensional perspective. The

Lorentz force in the rotation direction of both sides of the single bubble is equal in magnitude and opposite in direction. As the bubble grows, the maximum Lorentz force also increases. The Lorentz force distributions may have different dynamic effects on the rotational flow around the bubble and the growth behavior, which will be described in detail.

D. Rotational electrolyte flows

As mentioned above, the relatively higher Lorentz forces at the wedge-shaped areas can induce rotational electrolyte flows during the whole bubble growth cycle, as shown in Fig. 10. The black line represents the gas–liquid interface. It is interesting to note that the higher-velocity regions are located at oblique positions of the bubble equator and extend far into the bulk electrolyte, except for the rotational electrolyte flows both before and after the bubble detachment moment (about 3.89 s). It means that the higher rotational electrolyte flow velocities do not appear at the wedge-shaped areas with higher Lorentz forces as expected. One of the possible reasons is that the Lorentz forces induced rotational electrolyte flows may be constrained by the bottom wall of the horizontal microelectrode to satisfy the no-slip conditions. Another possibility is that the hydrogen bubble

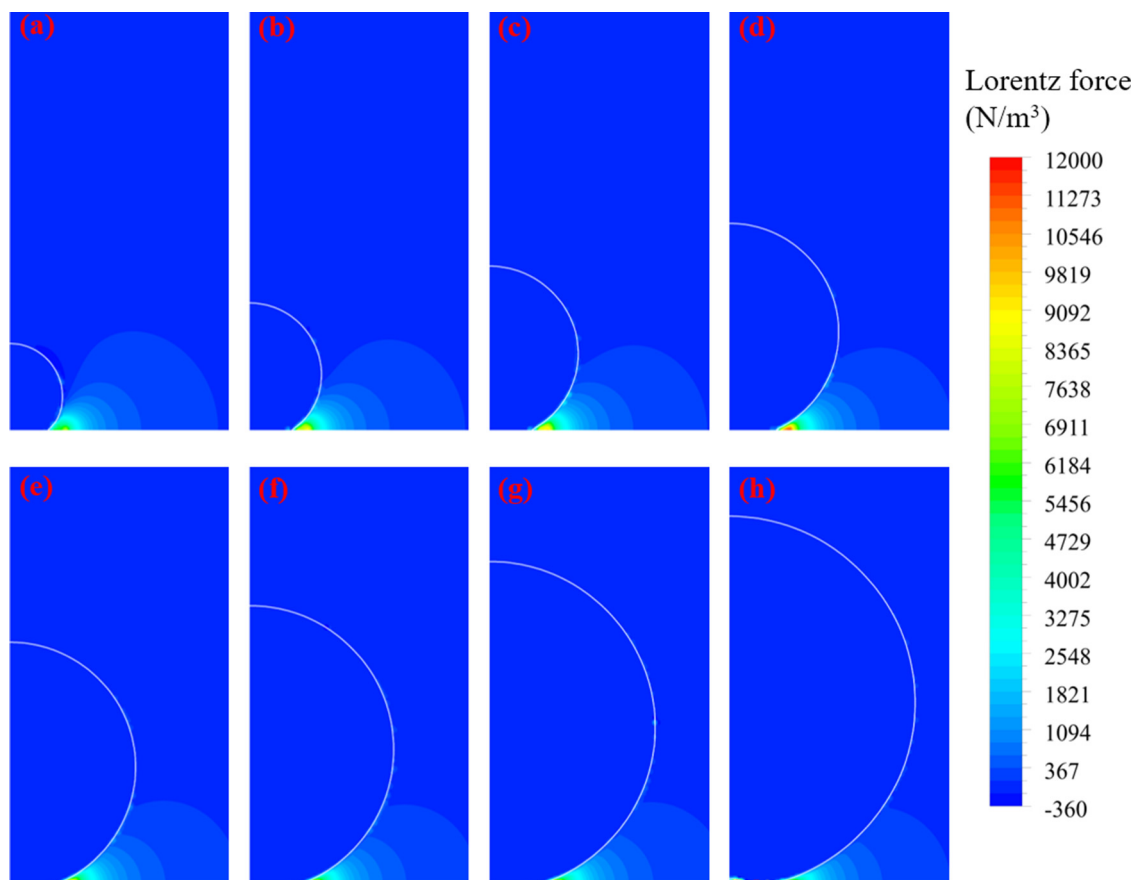


FIG. 9. Lorentz force distribution around a single hydrogen bubble on the horizontal microelectrode surface at different time instances: (a) 0.05, (b) 0.2, (c) 0.4, (d) 0.8, (e) 1.2, (f) 1.8, (g) 2.7, and (h) 3.89 s.

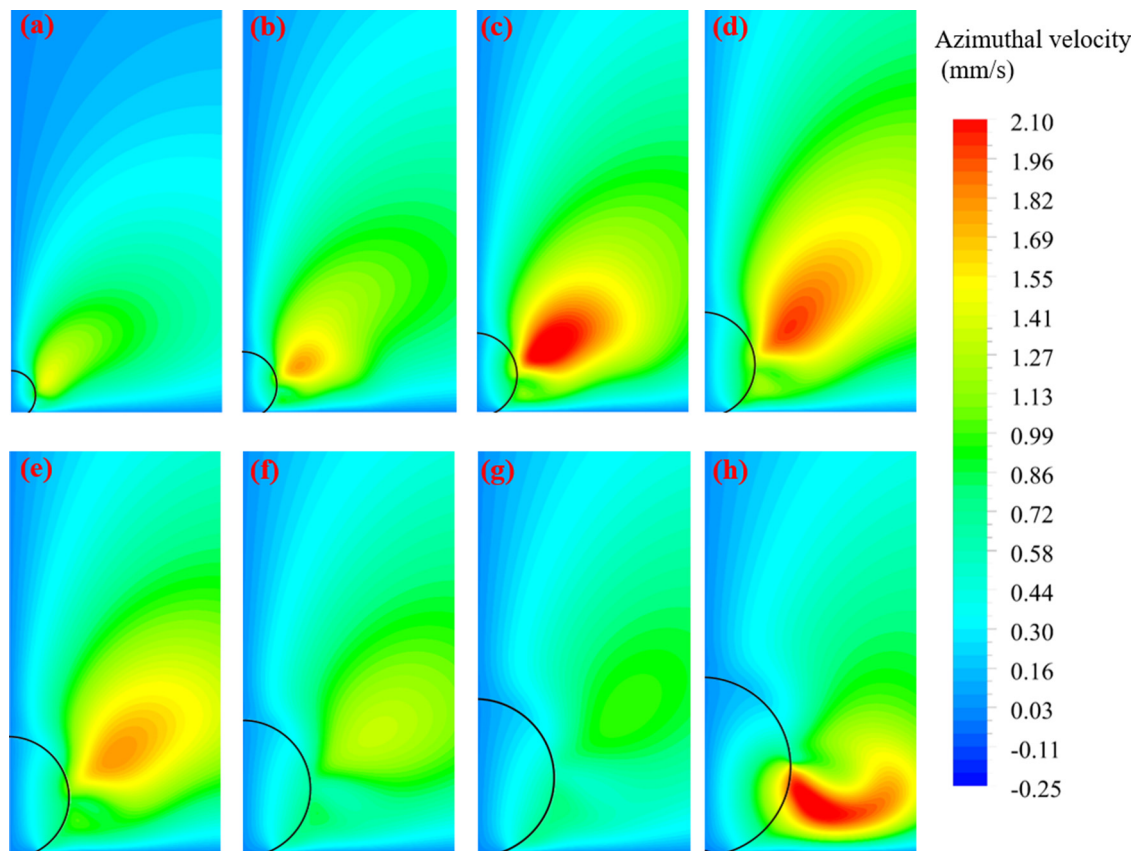


FIG. 10. The rotational electrolyte flow distribution around a single hydrogen bubble on the horizontal microelectrode surface at different time instances: (a) 0.05, (b) 0.2, (c) 0.4, (d) 0.8, (e) 1.2, (f) 1.8, (g) 2.7, and (h) 3.89 s.

expansion due to the continuous growth process may play an important performance. As the bubble grows, the maximum rotational electrolyte flow velocity and the corresponding higher-velocity regions are increased and then decreased. The rotational electrolyte flow velocity fields are more uniform in the later stages of the bubble growth cycle, as illustrated in Fig. 10(g), for example. However, the rotational electrolyte flows are significantly more complex when the bubble just detaches from the horizontal microelectrode surface.

In order to deeply reveal the impact of the bubble detachment process on the rotational electrolyte flow features, the dynamic evolutions of the rotational electrolyte flows for several successive time instances are illustrated in Fig. 11. The bubble detachment moment is approximately at around 3.89 s in the numerical simulation. It is evident from Figs. 11(b) and 11(c) that the bubble momentum has suddenly shifted within a short period of time of about 0.01 s (from 3.86 to 3.87 s), causing a significant high flow velocity feature. One possibility is that the bubble begins to detach from the horizontal microelectrode surface and it does not completely attach to the microelectrode. There is a sudden change in the electric field distribution around the bubble. This would cause high values of current density and Lorentz force as well as high flow velocity distribution at some special location. Then, the bubble detachment behavior would soon take place as

shown in Figs. 11(c)–11(e), which forces the bulk electrolyte around the bubble to gain energy source. As a consequence, the rotational electrolyte flow velocity increases gradually and extends far into the bulk to produce some wake vortices. However, when the bubble has already detached from the microelectrode surface [see Figs. 11(f)–11(h)], the rotational electrolyte flow is gradually decreased and then remains at a certain level.

IV. CONCLUSIONS

The numerical simulation of the dynamic growth process of a single hydrogen bubble on the horizontal microelectrode surface in the electrode-normal magnetic field is investigated by using the VOF method. Two typical mass transfer models considering the supersaturated dissolved hydrogen components from the liquid microlayer and bubble interface are built. This work provides an in-depth analysis and explanation of the dissolved hydrogen concentration, current density, Lorentz force, and rotational electrolyte flow fields. The simulated bubble growth diameter variation with time on the horizontal microelectrode surface is in good agreement with the experimental results. The simulation results and the calculation methods are highly reliable. The supersaturated dissolved hydrogen concentration and the mass transfer source at the wedge-shaped areas adjacent to the bubble foot

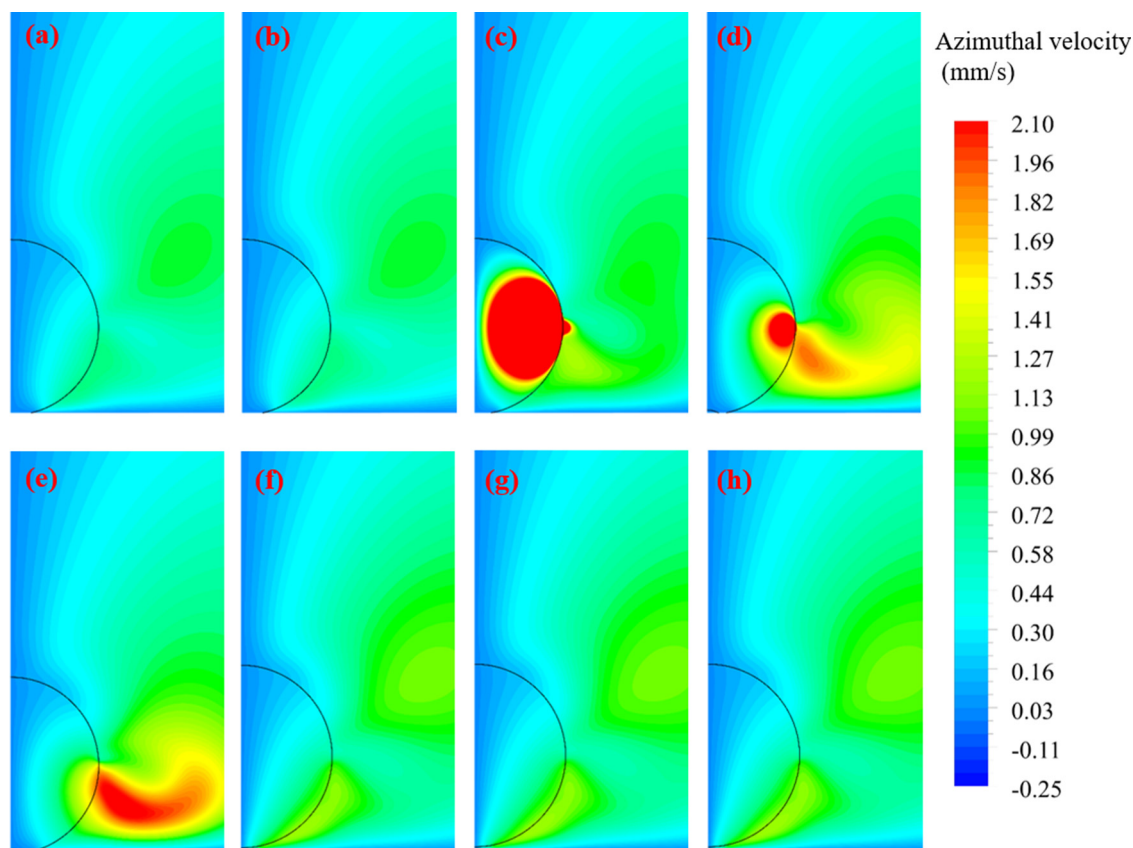


FIG. 11. The rotational electrolyte flow distribution around a single hydrogen bubble on the horizontal microelectrode surface before and after bubble detachment: (a) 3.85, (b) 3.86, (c) 3.87, (d) 3.88, (e) 3.89, (f) 3.90, (g) 3.92, and (h) 3.96 s.

appear significantly higher than those in the wider bulk bubble interface regions. The dissolved hydrogen mass transfer rate from the liquid microlayer beneath directly plays a dominant role. The current distortion and concentration effects through such a tiny microelectrode area and an insulating hydrogen bubble cause higher current density and Lorentz force distributions at the wedge-shaped areas, which become more pronounced when the bubble diameter increases. The higher rotational electrolyte flow velocities do not appear at the wedge-shaped areas with higher Lorentz forces as expected. The maximum rotational electrolyte flow velocity and the corresponding higher-velocity regions are increased and then decreased as the bubble grows. The bubble detachment behavior make the rotational electrolyte flows significantly more complex.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Shuiqing Zhan: Conceptualization (lead); Data curation (equal); Formal analysis (lead); Funding acquisition (lead); Investigation (equal); Methodology (equal); Software (equal); Validation (lead); Writing—review & editing (lead). **Rui Yuan:** Data curation (equal); Investigation (equal); Methodology (equal). **Yujie Huang:** Investigation (equal); Methodology (equal); Software (equal). **Wei Zhang:** Data curation (equal); Investigation (equal); Methodology (equal). **Bin Li:** Visualization (supporting); Validation (supporting); **Zhentao Wang:** Conceptualization (supporting); Investigation (equal). **Junfeng Wang:** Investigation (supporting); Project administration (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reason able request.

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